

B.Tech in Electrical Engineering

IIT Hyderabad — EE Department | 4-year programme · 2024 curriculum onwards

Semester 1 (August – December)

Code	Course Name	Credits	Type
BT1010	Introduction to Life Sciences	1	Basic Sciences
ID1063	Introduction to Programming	3	Basic Engg. Skills
EE1010	Mathematics for Electrical Engineers	3	Basic Sciences
EE1030	Matrix Theory	3	Basic Sciences
EE1101	Circuits and Network Analysis	3	Basic Engg. Skills
LA xxxx	English Communication	2	LA/CA
	Total	15	

Semester 2 (January – May)

Code	Course Name	Credits	Type
EE1060	Differential Equations and Transform Techniques	3	Basic Sciences
EE1204	Engineering Electromagnetics	2	Dept. Core
EE1201	Digital Systems	3	Dept. Core
EE1200	Electrical Circuits Lab	2	Lab Basket
EE1080	Probability	3	Basic Engg. Skills
EE1003	Scientific Programming for Electrical Engineers	2	Dept. Core
EE1501	Digital Systems Lab	1	Lab Basket
	Total	16	

Semester 3 (August – December)

Code	Course Name	Credits	Type
ID2230	Data Structures & Applications	3	Basic Engg. Skills
EE2000	Signal Processing	3	Dept. Core
EE2103	Semiconductor Device Fundamentals	3	Dept. Core
EE2200	Electrical Machines	3	Dept. Core
EE2080	Random Processes	1	Basic Engg. Skills
(Lab Basket)	Lab Basket	2	Lab Basket
(Sci Basket)	Science Basket	2	Basic Sciences
	Total	17	

Semester 4 (January – May)

Code	Course Name	Credits	Type
EE2400	Analog Electronics	3	Dept. Core
EE2600	Power System Analysis	3	Dept. Core
EE2802	Machine Learning	3	Basic Engg. Skills
EE2101	Control Systems	3	Dept. Core
(Sci Basket)	Science Basket	2	Basic Sciences
(Lab Basket)	Lab Basket	2	Lab Basket
EM3020	Introduction to Entrepreneurship	1	LA/CA
	Total	17	

Semester 5 (August – December)

Code	Course Name	Credits	Type
EE3301	Intro to VLSI Design	2	Dept. Core
EE3500	Machines Lab	2	Lab Basket
EE3501	Power Electronics	3	Dept. Core
EE3700	Communication Systems	3	Dept. Core
EE3004	Current Trends in EE	1	Dept. Core
(Sci Basket)	Science Basket	2	Basic Sciences
(Free Elec)	Free Elective	1	Free Elective
XX xxxx	LA/CA Elective	2	LA/CA
	Total	16	

Semester 6 (January – May)

Code	Course Name	Credits	Type
EE xxxx	Department Electives	12	Dept. Elective
EE xxxx	Lab Basket	2	Lab Basket
EE3014	Current Trends in EE	1	Dept. Core
LA xxxx	LA/CA Elective	2	LA/CA
	Total	17	

Semester 7 (August – December)

Code	Course Name	Credits	Type
EE xxxx	Department Electives	6	Dept. Elective
LA xxxx	LA/CA Electives	2	LA/CA
FE xxxx	Free Electives	6	Free Elective
	Total	14	

Semester 8 (January – May)

Code	Course Name	Credits	Type
EE xxxx	Department Electives	9	Dept. Elective
LA xxxx	Ethics and Values	1	LA/CA
FE xxxx	Free Electives	6	Free Elective
	Total	16	

Science Basket (6 credits required, suggested across Semesters 2–5)

Code	Course Name
EP1108	Modern Physics
CY1010	Environmental Chemistry
MA1230	Series of Functions
MA2130	Complex Variables
MA2120	Transform Techniques
MA1130	Vector Calculus
MA3060	Numerical Analysis
MA2150	Introduction to Metric Spaces
EP2200*	Thermodynamics
EP2108*	Special Relativity
EP2228*	Fluid Dynamics
EP2100*	Classical Mechanics
CC1010*	Introduction to Climate Change
MA1240*	Combinatorics
MA2140*	Introduction to Statistics
MA2510*	Linear Programming
EP3100	Quantum Mechanics 1
EE3210	Quantum Mechanics 2
EP3200	Solid State Physics
EP3120	Statistical Mechanics
CY1031	Chemistry Lab
EP1031	Physics Lab

Lab Basket (minimum 6 credits)

Code	Course Name	Credits
EE2401	Analog Lab	2
EE3401	Microprocessor Lab	2
EE3701	Comm Lab	2
EE2801	DSP Lab	2
EE3802	FPGA Lab	2
EE3011	Electromagnetics and Applications Lab	2
EE2501	Semiconductor Devices Lab	2
EE2301	Electronic Devices and Circuits Lab	2
EE2504	Electronic System Design Project Lab	2

Optional Independent Projects

Code	Course Name
EE3005	Independent Project
EE3025	Independent Project
EE4015	Independent Project
EE4025	Independent Project

Note: This curriculum applies to students admitted from 2024–25 onwards. Students admitted between 2020 and 2023 follow the 2020–2023 curriculum.